

COMP9444 Project Summary

Unpaired Image to Image Translation on Video Game Graphics

by some group

Melanie Daixing (z5419027), Tristan Fischer (z5310974), Calvin Liu (z5367106), LeoXu(z5361413),GuozhiZhao(z5319307)

I. Introduction

The visual style of older games can often appear dated and low in quality due to the technical limits of computer hardware being unable to create high-end visual scenery and graphics at the time. As a result, many older games often feature pixelated textures, low-resolution assets, simplified visual effects and a dimmer, less vibrant design. Now, with rapid advancements in technology, player expectations for a visually immersive gaming experience have risen.

The aim of our project is to utilise the increase in computational power, especially the powerful GPUs, in modern gaming devices to successfully execute style transfer techniques efficiently. This involves transferring the visual style of the modern game Valorant and adapting the style to the context of the older game, Counter Strike 1.6. Specifically, this will include adapting the lighting, textures and colour schemes to align with the visual aesthetic of more contemporary games.

We wish to explore and evaluate some of the different methods for transferring image styles, and test whether these methods can be used to transfer the visual style of modern games to older games of a similar genre, giving them new, visually appealing graphics. Testing the transfer of visual styles between games of a similar aspect is crucial, as it ensures that the adapted visuals will still align with the overall theme and elements of the original game, creating a more cohesive experience for players.

II. Literature Review

Generative Adversarial Networks (GANs)

GANs utilise an adversarial framework, where a generative model is trained simultaneously alongside a discriminative model that learns to determine whether data provided to it is from the training data or from the generative model. The generative model learns to increase the quality of the data it creates such that the discriminative model becomes unable to differentiate whether it is real or not. GANs have made many breakthroughs in image generation and many innovative image generation methods have been developed using GANs.

Unpaired Image-to-Image Translation

Image-to-image translation aims to learn a mapping function between two sets of images that share similar characteristics however utilise different styles. However given two marginal distributions, as there can be an infinite number of joint distributions that can lead to the given marginal distributions, it is impossible to gain any information from this. To solve this problem, assumptions are made by different methods of image translation that all attempt to provide valid and efficient solutions for vision and graphics tasks.

Cycle Consistency

Many methods utilise cycle consistency, which exploits the transitive property and learns an inverse mapping from the output domain and checks if the input can be reconstructed from the output. This is a one-to-one method that introduces a cycle consistency loss to push the generative networks to be

consistent with each other. However, due to its restrictive nature it is not suitable for all types of tasks, especially tasks where images from one domain contain additional information compared to the other.

Contrastive Learning

This method only needs to learn the mapping in one direction, simplifying the training process. The discriminator is not used and the generator function is separated into an encoder and a decoder, to produce an output image. Adversarial loss is introduced to ensure that the output is visually similar to images from the target domain. This learning method is used on both a whole image and a patch level so that the mutual information in corresponding patches between input and output images is maximised.

Density Changing Regularisation

The assumption that this method uses is that image patches of high probability density in the input domain should also be mapped to a patch with high density in the output. To ensure that the images created by the generator look like images in the target domain, adversarial loss and identity mapping loss are introduced to regularise the mapping functions. Alongside this, variances in density changes are penalised using density estimators so that the optimal mapping can be more easily found.

Unpaired Neural Schrödinger Bridge

This is an extremely new method that applies diffusion models, which are capable of high quality data synthesis, combined with a Schrödinger Bridge based approach to translation tasks. This method also utilises an advanced Markovian discriminator, which is able to distinguish data on a patch level, and is extremely effective at capturing the style of the target data. The multi-step nature of this model allows it to have better performance than GAN based translation methods.

Methods

We wished to compare the accuracy and effectiveness of a variety of image style transfer methods to enhance the visual style of Counter Strike 1.6 (2003) so that it takes on the style of Valorant (2020). The four models we chose to train, test and evaluate were CycleGAN, Contrastive Unpaired Translation (CUT) GAN, Unpaired Image-to-Image Translation with Density Changing Regularization (DECENT) and Unpaired Image-to-Image Translation via Neural Schrödinger Bridge (UNSB). Our project did not use pre-trained models as we created our own dataset, and trained the models on the dataset that we made.

III. Experimental Setup

Our dataset was handmade from recorded Counter Strike 1.6 and Valorant footage and contains approximately 4000 images for each game. We split our dataset into 4000 images for each game used in training, and 4000 images for each game used in testing. The data includes both gameplay and environment footage of different maps with different visual elements such as environment textures and colourings, giving our models the chance to learn and apply the changes.

We chose our model parameters based off of the CycleGAN paper, which had a dataset comprising roughly 1000 images for each class. This paper trained their dataset for 100 epochs with the same learning rate and then decayed their learning rate for the next 100 epochs. As our dataset is approximately 4 times larger, we decided to scale our epoch number by $\frac{1}{4}$ and due to our time constraints, we further lowered the number of epochs to 20. We trained each model with a batch size of 4 except UNSB which we trained with a batch size of 1. CycleGAN took 10 minutes per epoch to train, CUT GAN took 2 minutes, DECENT took 7.5 minutes and UNSB took 33 minutes to train.

IV. Results

Quantitative Results

Fréchet Inception Distance/Kernel Inception Distance

CycleGAN was evaluated as holding the best performance through both FID/KID and Decent received the worst performance. This is consistent with qualitative evaluations. However a key issue with this methodology is that, while CycleGAN is the best in terms of visual quality, this can largely be attributed to the less extensive modifications compared to the other models and a departure from the intention of the project. Evaluating authenticity and ensuring the model is generative is an open-question with no standard quantitative metric available.

Fréchet Video Distance

The data was evaluated through FVD using three distinct metrics: FVD2048_16F (FVD with 16 frames), FVD2048_128F (FVD with 128 frames) and FVD2048_128F_ss8F (FVD with 16 frames, sampled with 8-frames interval). Through these metrics, the most consistent model at high performance is CycleGAN. This is likely due to CycleGAN's property of cycle consistency, an assumption the relationship between two domains is a bijection, which given the current dataset allows for improved temporal consistency across frames.

Qualitative Results



The original images provided include one snapshot of gameplay and two snapshots of the environment in varying light and dark gradients. The main objects included are the sky, walls, the ground and obstacle crates.

In terms of colour palette, the original image is tinted a dark yellow, with shadowed areas being extremely dark. The images generated by CycleGAN and CUT have an unrealistic deep orange tint, with artificial lighting replacing the shadows, more so for CUT than CycleGAN. On the other hand, DECENT and UNSB produce a slightly more peach coloured reshading of the images however they also replace the shadows with artificial brightness.

The textures in the original images make the ground seem more cobblestone like, and the lines are mostly well defined. All the models have undergone varying amounts of texture smoothing in relation to the ground and walls, with CycleGAN still maintaining some line definition for the cobblestone textures and CUT, DECENT and UNSB have almost no fine detailing in relation to ground textures. In all of the output images, the map on the top left is randomly generated and doesn't match the gameplay as Counter Strike 1.6 does not have a map and Valorant does.

Due to the variety in gun shapes in our dataset, the style transfer of the gun caused many issues in the output images of some models. CycleGAN transfers the gun skin the best, keeping the model of gun consistent and well outlined and just changing the colour of the gun model to involve patterns. CUT and UNSB both had some issues with transferring the model of the gun, as their images include different gun models, however they mostly transfer the colouring well with only some slight blurring around the edges of the gun. The worst performing model in this element was DECENT, who had a lot of trouble keeping the gun model consistent which may be due to its style transfer process having image density as an important factor.

In terms of overall image coherence, CycleGAN by far produces the best images, with CUT performing moderately, UNSB slightly worse and DECENT performing poorly. However, DECENT also attempts to transfer more of the modern game style than the other methods, while the other models mainly did a colour reshading, which may be why it performed so poorly. This may also be due to the models' respective style transfer methods, as CUT GAN, DECENT and UNSB all utilise methods like density estimations and patchwise contrastive learning that preserve structural similarities during translation. This has also caused the minimap textures, which may be mistaken as having a similar texture to the environment, for the results of these models to appear in the middle of the screen. The gun textures also have not transferred well for these three models, with the different colourings and shapes of Valorant guns causing the guns in the output image to become misshapen.

V. Conclusions

Improvements to the dataset could be made by recording the environment and the gameplay of Valorant with the minimap disabled. Counter Strike 1.6 gun types also look mostly similar to each other, however Valorant gun types do not. Due to the Valorant dataset having recorded footage with different types of guns, the output images for some of the models sometimes replaced the intended larger rifle with an unintended smaller pistol.

The larger, rifle gun type should also be consistently used in all the recordings so that the models are able to consistently transfer the gun skin textures.

Without a time constraint, all of the models could be trained for a longer time by increasing the number of epochs or adjusting other parameters such as the training rate, which may also aid in obtaining a more accurate image transfer.

Reflecting on the project our group has worked on this term, there are many ways we can improve and further develop our work. This includes finding a way to stylize older games to look like modern games in real time. This would require a neural network capable of extremely fast processing, with low latency and an ability to sustain a moderate number of frames per second.

Potential challenges would be finding a base image style transfer model that is able to accurately output translated images, and then adjusting the parameters and potentially the model itself to improve performance to a playable quality.

VI. References

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